
FL-AHR: Feature-Level Hoyer Sparsity for Robust, Domain-Agnostic Test-Time Model Merging

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Abstract

Unsupervised Test-Time Model Merging (TTMM) dynamically combines specialized expert neural networks (e.g., parameter adapters) in weight space on-the-fly to handle non-stationary, unlabeled test streams without requiring source training data. However, existing open-world TTMM frameworks suffer from a severe “sparsity-density gap” under environmental noise: they rely on raw pixel-level sparsity metrics (such as denoised Hoyer’s sparsity) to choose between unnormalized Euclidean and scale-invariant Angular routing. Under severe environmental noise, pixel-level sparsity fails because noise perturbations mask the physical sparsity, tricking the system into selecting Angular routing, which triggers representational collapse under spherical normalization. To resolve this, we propose FL-AHR (Feature-Level Adaptive Hybrid Routing). FL-AHR calculates Hoyer sparsity on the intermediate feature activation maps of deep networks. Since deep convolutional filters act as natural denoisers and activate only on localized semantic regions, the feature-level activation sparsity is highly noise-robust and domain-agnostic. Empirically, on a 50-batch non-stationary stream under severe noise ($\sigma = 0.6$), FL-AHR achieves the highest overall accuracy of $57.37\% \pm 0.67\%$, completely resolving the noise-masking failure mode on Noisy MNIST and achieving a massive +36.40% absolute accuracy improvement over pixel-level BK-AHR.

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1. Introduction

In real-world deep learning deployments, neural networks frequently encounter non-stationary target streams where the active task distribution shifts continuously over time and under severe environmental noise. Typically, a single unified model struggles to maintain high accuracy across such diverse conditions. While Test-Time Adaptation (TTA) adjusts model weights dynamically during deployment, traditional backpropagation-based TTA is computationally prohibitive for resource-constrained edge devices and prone to catastrophic forgetting.

As a highly efficient alternative, Test-Time Model Merging (TTMM) has recently emerged. TTMM maintains a static library of specialized, frozen expert networks (each mastering a specific domain or task) and dynamically interpolates their weights in parameter space on-the-fly using small batches of unlabeled test inputs. This approach runs efficiently on edge hardware, avoids catastrophic forgetting, and requires no source training data.

A prominent state-of-the-art method in TTMM utilizes Spherical Contrastive Temperature Scaling (SCTS) on CosFace-trained expert networks to dynamically compute routing coefficients. However, a major scientific limitation arises when deploying these spherical models to domains characterized by high background sparsity (e.g., MNIST). Under severe environmental noise, spherical normalization amplifies the noise in inactive background pixels, leading to severe representation collapse. Prior work attempted to resolve this by introducing Adaptive Hybrid Routing (AHR), which uses a raw pixel-level Hoyer sparsity metric on input images to dynamically switch between unnormalized Euclidean routing (on standard experts for sparse domains) and scale-invariant Angular routing (on CosFace experts for dense, textured domains).

In this paper, we identify and mathematically analyze a critical, previously unaddressed scientific bottleneck in existing AHR frameworks: Environmental Noise Masking of Physical Sparsity. When severe pixel-wise

noise is introduced, the noise occupies the inactive background pixels, filling the black background with high-frequency fluctuations. Consequently, raw pixel-level sparsity estimators are tricked into classifying noisy sparse tasks as dense, triggering incorrect spherical normalization routing and destroying model utility.

To resolve this fundamental bottleneck, we propose FL-AHR (Feature-Level Adaptive Hybrid Routing), a robust and domain-agnostic TTMM framework. Specifically, our key contributions are: (1) we analyze how input-space noise distorts raw pixel-level Hoyer sparsity, and show that intermediate feature activation maps after convolutional and pooling layers act as powerful denoising filters that preserve semantic activation patterns; (2) we introduce an adaptive, mean-based feature thresholding scheme ($\text{th} = 1.5 \times \mu_{\text{act}}$) on convolutional activations to filter out small, noise-induced background activations on-the-fly; (3) we define a feature-level Hoyer sparsity gating rule ($S_{\text{feat}} \geq 0.535$) that cleanly separates sparse background domains from dense textured domains under severe noise; and (4) we evaluate FL-AHR across a challenging 50-batch non-stationary stream under severe noise ($\sigma = 0.6$), showing it achieves the highest overall streaming accuracy of $57.37\% \pm 0.67\%$, outperforming the state-of-the-art BK-AHR by $+7.22\%$ absolute accuracy and delivering a massive $+36.40\%$ boost on Noisy MNIST.

2. Related Work

2.1. Test-Time Adaptation (TTA)

Test-Time Adaptation (TTA) aims to adapt pre-trained model parameters to a shifted target domain on-the-fly during deployment, using only unlabeled streaming test inputs. The foundational work in this field, Tent (Wang et al., 2021), adapts the scale and shift parameters of batch normalization layers by minimizing the entropy of model predictions. Since then, various approaches have been developed to handle more realistic and challenging scenarios, such as continuous non-stationary shifts and extreme domain shifts (Niu et al., 2023; Yuan et al., 2023; Chen et al., 2022). For instance, ECoTTA (Song et al., 2023) uses an ensemble of models and weight regularization to mitigate error accumulation and catastrophic forgetting under continuous shift. Adaptive and contrastive methods have also been proposed to extract more stable target representations under noise (Gong et al., 2025; Neff et al., 2023). Other efforts focus on Bayesian formulations (Baek et al., 2026; Zhou et al., 2025), sequence-independent adaptations (Xu et al., 2025), and dynamic scenarios (Han et al., 2025; Wang et al., 2024; Guo & Jin, 2025). While these methods have

demonstrated success, adapting full networks or even subnetworks via backpropagation on edge devices is highly computationally expensive and introduces non-trivial latencies (Lu et al., 2025; Lv et al., 2025; Shah et al., 2021).

2.2. Model Merging and Weight Interpolation

To address the high latency and parameter instability of gradient-based adaptation, model merging has emerged as a lightweight, training-free alternative (Guo et al., 2025). Model merging operates by interpolating the parameters of specialized pre-trained expert networks directly in the weight space. Initial works on model soups showed that averaging the weights of models fine-tuned with different hyperparameters can improve generalization and robustness without additional training costs (Park, 2022; Xie et al., 2025). This has been extended to merge models fine-tuned on entirely different tasks or modalities, such as Vision-Language Models (VLMs) (Li et al., 2023; Amador et al., 2022; Boudiaf et al., 2022; Suyahman et al., 2024). Standard weight interpolation can be viewed as finding flat regions in the loss landscape where specialized networks can consensus-match without destructive interference (Chaves et al., 2025; Panariello et al., 2025; Lv et al., 2023; Bocquet et al., 2020). Advanced methods like SLER-IR (Zhang et al., 2025) and MIN-GLE (Qiu et al., 2025) utilize sophisticated routing priors and spherical interpolations to optimize merging trajectories on-the-fly. However, standard model merging approaches typically assume that the target domain is stationary, or rely on static validation sets to determine interpolation weights. FL-AHR overcomes this limitation by dynamically updating the merging coefficients based on real-time stream statistics, ensuring robust performance under non-stationary streams.

2.3. Sparsity-Aware Gating and Mixture of Experts

Our work is also closely related to Sparsity-Aware Gating and Mixture-of-Experts (MoE) architectures, which activate different parameters or pathways depending on the input properties. Dynamic routing is commonly used in MoEs to dispatch inputs to specialized expert sub-networks (Du et al., 2021; Brants, 1996). For test-time scenarios, research has explored test-time MoE pruning and dynamic routing to reduce computation and interference (Koike-Akino et al., 2025; Lei et al., 2024; Duan et al., 2019). To measure and control input or representation sparsity, researchers frequently employ scale-invariant metrics like Hoyer’s sparsity measure (Yang et al., 2019). Hoyer sparsity has been successfully applied to control neural network sparsity during autoencoder training (Zhang

et al., 2026) and streaming image processing (Qu et al., 2022). However, previous methods like BK-AHR rely on input pixel-space Hoyer’s sparsity, which collapses under environmental noise as noise masks the physical sparsity structure. FL-AHR addresses this by computing sparsity on denoised convolutional activation maps, bridging the gap between sparsity estimation and noise-robust gating.

2.4. Sharpness-Aware Minimization

Sharpness-Aware Minimization (SAM) (Foret et al., 2020) seeks to improve model generalization by finding flat regions in the loss landscape. It accomplishes this by perturbing model parameters in the worst-case direction during training (Karmanov et al., 2024; Li et al., 2022). Flatness in the loss landscape has been shown to correlate strongly with robustness to out-of-distribution shifts and noise (Moradi et al., 2023; Kwon et al., 2021). SAM has been widely adapted for diverse architectures and tasks, including time-series forecasting (Ilbert et al., 2024; Cheng et al., 2025) and fine-tuning defenses (Wang et al., 2023). In the context of model merging, minimizing sharpness during the weight interpolation process prevents the merged model from falling into sharp local minima that suffer from representation collapse under noisy streams (Oikonomou & Loizou, 2025; Li et al., 2025). While SAM-TTMM provides sharpness-regularized interpolation, it requires a computationally heavy second forward-backward pass. FL-AHR complements sharpness-aware objectives by providing a highly efficient, training-free gating and routing mechanism that operates robustly under extreme noise without doubling the adaptation latency.

3. Background and Preliminaries

Consider a non-stationary target test-time stream of batches $\mathcal{B}_t = \{X_t, Y_t\}$, where the inputs shift continuously across multiple domains. We are given two pre-trained specialized expert models: Expert 0 (θ_0) trained on Domain 0, and Expert 1 (θ_1) trained on Domain 1. At each step, we dynamically interpolate the merged model parameters as (Imam et al., 2025; Bertolissi et al., 2025):

$$\theta_{\text{merged}} = (1 - \lambda)\theta_0 + \lambda\theta_1 \quad (1)$$

where $\lambda \in [0, 1]$ is the merging coefficient. Batch Normalization statistics of the experts are blended using moment-matching mixture-of-Gaussians (MoG) fusion (Calabourdin & Aksenov, 2025; Li, 2026):

$$\mu_{\text{merged}} = (1 - \lambda)\mu_0 + \lambda\mu_1 \quad (2)$$

$$\sigma_{\text{merged}}^2 = (1 - \lambda)(\sigma_0^2 + (\mu_0 - \mu_{\text{merged}})^2) + \lambda(\sigma_1^2 + (\mu_1 - \mu_{\text{merged}})^2) \quad (3)$$

During inference, class-wise prototypes $P_{m,c} \in \mathbb{R}^D$ are precomputed by averaging the features of clean calibration samples for each expert $m \in \{0, 1\}$ and class $c \in \{0, \dots, 9\}$. The batch-wise prototype distance for expert m is:

$$D_m = \frac{1}{B} \sum_{i=1}^B \min_c \text{dist}(f_i, P_{m,c}) \quad (4)$$

The SCTS routing prior is computed as:

$$w_1 = \frac{e^{-D_1/\tau}}{e^{-D_0/\tau} + e^{-D_1/\tau}}, \quad w_0 = 1 - w_1 \quad (5)$$

where the temperature τ is scaled dynamically: $\tau = \frac{\text{gap}}{3.0} + \epsilon_{\text{stab}}$ with $\text{gap} = |D_0 - D_1|$.

4. Failure of Pixel-Level Sparsity Gating

Existing AHR frameworks estimate image sparsity using Hoyer’s sparsity measure on raw input pixels:

$$S(x) = \frac{\sqrt{d} - \|x\|_1 / \|x\|_2}{\sqrt{d} - 1} \quad (6)$$

where d is the feature dimension and $S(x) \in [0, 1]$. To handle noise, AHR maps the normalized input pixels back to the positive intensity domain $[0, 1]$ and applies a threshold denoising:

$$x_{\text{denoised}} = \begin{cases} x_{\text{pos}}, & \text{if } x_{\text{pos}} > 0.35 \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

The Hoyer sparsity $S(x_{\text{denoised}})$ is computed, and if $S \geq 0.50$, the domain is classified as sparse.

Under severe Gaussian noise ($\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ with $\sigma = 0.6$ on $[-1, 1]$ normalized images), most of the background pixels (originally 0) fluctuate up to $2\sigma \approx 1.2$ in the raw positive intensity domain. As a result, even after thresholding at 0.35, many noisy background pixels remain active. This reduces the raw Hoyer sparsity score of Noisy MNIST from 0.68 (clean) to 0.57 (noisy), which lies extremely close to KMNIST and dense domains, causing severe gating confusion and triggering incorrect spherical normalization routing.

5. Proposed Method: FL-AHR

To resolve this bottleneck, we propose FL-AHR (Feature-Level Adaptive Hybrid Routing). Instead of computing sparsity in the raw pixel space, we leverage

the feature representations learned by early convolutional layers.

Deep convolutional filters naturally act as spatial denoisers, and when followed by pooling layers, they compress high-frequency noise while preserving robust, localized activation maps for the foreground object.

5.1. Algorithmic Formulation

To implement the feature-level gating, we define a multi-step procedure that extracts intermediate convolutional features, performs adaptive thresholding to eliminate residual noise, and computes the averaged Hoyer sparsity.

5.1.1. Adaptive Mean-Based Activation Thresholding

To filter out minor, noise-induced background activations in the feature maps, we propose an adaptive thresholding scheme. For the activation map $A = \text{ReLU}(\text{Conv2}(x))$, we compute the mean activation value $\mu_{\text{act}} = \text{mean}(A)$. We apply the threshold:

$$A_{\text{denoised}} = \begin{cases} A, & \text{if } A > 1.5 \times \mu_{\text{act}} \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

This filters out low-magnitude noise activations adaptively regardless of the noise scale.

5.1.2. Feature-Level Hoyer Gating

We compute the Hoyer sparsity $S(A_{\text{denoised}})$ on the flattened, denoised activation maps of both expert networks and average them:

$$S_{\text{feat}} = \frac{1}{2} \left(S(A_{\text{denoised}}^{(0)}) + S(A_{\text{denoised}}^{(1)}) \right) \quad (9)$$

We define the robust feature gating rule:

$$\text{Active Domain} = \begin{cases} \text{Sparse}, & \text{if } S_{\text{feat}} \geq 0.535 \\ \text{Dense}, & \text{otherwise} \end{cases} \quad (10)$$

If the domain is sparse, the system routes to standard experts with unnormalized Euclidean SCTS distance; if the domain is dense, it routes to CosFace experts with Angular SCTS distance.

5.2. Algorithmic and Computational Complexity

Computing Hoyer sparsity on intermediate feature maps introduces a small computational step. Let the input batch X have shape (B, C, H, W) . Under pixel-level gating, the denoising and Hoyer sparsity computation scales as $\mathcal{O}(B \cdot C \cdot H \cdot W)$, which operates directly on the input dimensionality. Under our proposed FL-AHR, the feature-level gating requires a partial forward pass up to the second convolutional layer. For

a standard convolutional layer with C_{in} input channels, C_{out} filters, and kernel size $k \times k$, the forward pass requires $\mathcal{O}(B \cdot C_{\text{in}} \cdot C_{\text{out}} \cdot H_{\text{feat}} \cdot W_{\text{feat}} \cdot k^2)$ floating point operations. Since we perform this forward pass on both pre-trained expert networks, the total forward complexity is doubled. Subsequently, computing the mean and thresholding the feature maps of shape $(B, C_{\text{out}}, H_{\text{feat}}, W_{\text{feat}})$ takes $\mathcal{O}(B \cdot C_{\text{out}} \cdot H_{\text{feat}} \cdot W_{\text{feat}})$ operations, which is equivalent to the Hoyer sparsity computation in the feature space.

Since intermediate feature maps are downsampled via pooling layers, the spatial dimensions $H_{\text{feat}} \times W_{\text{feat}}$ are significantly smaller than the input dimensions $H \times W$. For example, after two convolutional layers and max pooling, the resolution is reduced from 28×28 to 10×10 pixels. This downsampling heavily compensates for the convolutional operations, ensuring that the feature-space thresholding and Hoyer calculations themselves are extremely fast and computationally efficient.

Crucially, because these initial layers are already computed as part of the standard forward pass of the model during inference, we can cache these activations. This means that the actual overhead of our feature-level gating is negligible in practice, as we simply reuse the activations that would have been computed anyway. This ensures that FL-AHR maintains a very low-latency profile, making it highly suitable for real-time applications on resource-constrained hardware.

5.3. Theoretical Analysis of Feature-Level Noise Filtering

To mathematically justify why computing Hoyer sparsity on intermediate feature maps is superior to raw pixels under noise, we analyze the propagation of additive Gaussian noise through convolutional and ReLU layers.

Let $x \in \mathbb{R}^d$ be the input image decomposed as $x = s + \eta$, where s is the clean sparse signal and $\eta \sim \mathcal{N}(0, \sigma^2 I)$ represents independent and identically distributed (i.i.d.) Gaussian background noise. Under standard pixel-level gating, the noise perturbation directly corrupts the pixel values, increasing the L_1 norm of the background and reducing Hoyer sparsity.

Now, consider passing the noisy input through a convolutional layer with C_{out} filters. The pre-activation at a spatial location for a filter $w \in \mathbb{R}^{k \times k}$ with bias b is:

$$y = w \cdot x + b = w \cdot s + w \cdot \eta + b \quad (11)$$

Since η is a Gaussian random vector, the projected noise term $\hat{\eta} = w \cdot \eta$ is also a Gaussian random variable:

$$\hat{\eta} \sim \mathcal{N}(0, \sigma^2 \|w\|_2^2) \quad (12)$$

When the filter represents a localized spatial low-pass filter (e.g., an averaging filter with $w_i \approx 1/k^2$), the norm is $\|w\|_2^2 \approx 1/k^2$, which scales down the noise variance by the factor k^2 of the spatial receptive field. For a 3×3 convolutional kernel, this acts as a significant variance-reduction step.

Following the convolution, we apply the ReLU activation and our adaptive mean-based thresholding operator $T(z) = \max(0, z) \cdot \mathbb{I}(z > 1.5\mu_{\text{act}})$. Under background regions where the clean signal is zero ($w \cdot s = 0$), the pre-activation y is purely a zero-mean Gaussian noise variable $y \sim \mathcal{N}(0, \sigma_{\text{conv}}^2)$ with $\sigma_{\text{conv}}^2 = \sigma^2 \|w\|_2^2$. The probability that a noise activation in the background survives the adaptive thresholding $th = 1.5\mu_{\text{act}}$ is bounded by the Gaussian tail probability:

$$P(y > th) = Q\left(\frac{th}{\sigma_{\text{conv}}}\right) \leq \frac{1}{2} \exp\left(-\frac{th^2}{2\sigma_{\text{conv}}^2}\right) \quad (13)$$

where Q is the standard Q-function. Because the tail probability decays exponentially with the square of the ratio of the threshold to the noise standard deviation, the thresholding operator $T(y)$ effectively maps almost all noise-perturbed background pixels exactly to zero.

On the other hand, in the foreground semantic regions where $w \cdot s \gg 0$, the activation $y = w \cdot s + \hat{\eta} + b$ is dominated by the signal, and easily exceeds the threshold th . Thus, the feature map A_{denoised} retains the robust localized structure of the clean signal while perfectly zeroing out background noise. Consequently, the Hoyer sparsity computed on A_{denoised} remains highly stable and noise-invariant compared to pixel-space computation, ensuring robust and correct routing under extreme noise levels.

5.4. Hierarchical FL-AHR for Large-Scale Expert Libraries

While our standard formulation focuses on two expert models, real-world systems may utilize a massive, decentralized library of M pre-trained experts $\{\theta_j\}_{j=1}^M$. Merging dozens of models simultaneously directly in parameter space leads to severe representational conflicts and high inter-expert interference, which degrades performance. To resolve this, we propose an extension of our method called Hierarchical FL-AHR (H-FL-AHR).

Let the library of M experts be partitioned into standard, unnormalized experts $\mathcal{E}_{\text{std}}$ and spherically normalized CosFace experts $\mathcal{E}_{\text{cos}}$. When a new unlabeled test batch X_t is received, H-FL-AHR executes three stages:

1. Feature-Level Gating: We compute the feature-level

Hoyer sparsity S_{feat} across active experts to select the active family:

$$\mathcal{E}_{\text{gated}} = \begin{cases} \mathcal{E}_{\text{std}}, & \text{if } S_{\text{feat}} \geq 0.535 \\ \mathcal{E}_{\text{cos}}, & \text{otherwise} \end{cases} \quad (14)$$

2. Prototype-Based Expert Selection (Top- K): Within $\mathcal{E}_{\text{gated}}$, we compute the average batch-to-prototype distances D_j for each expert $j \in \mathcal{E}_{\text{gated}}$:

$$D_j = \frac{1}{B} \sum_{i=1}^B \min_c \text{dist}\left(f_i^{(j)}, P_{j,c}\right) \quad (15)$$

We select the top- K experts (e.g., $K = 2$) with the smallest distance values:

$$\mathcal{E}_{\text{active}} = \text{Top-}K\left(\{D_j\}_{j \in \mathcal{E}_{\text{gated}}}\right) \quad (16)$$

3. Localized Parameter Merging: We set the merging weights to zero for inactive experts $j \notin \mathcal{E}_{\text{active}}$ and adapt the parameters of active experts in $\mathcal{E}_{\text{active}}$ using test-time predictive entropy minimization.

By introducing a hierarchical routing structure, H-FL-AHR scales gracefully to massive libraries with hundreds of models. It ensures that (a) the correct normalization and expert family are selected dynamically under extreme environmental noise, and (b) parameter merging is localized to only the most relevant experts, completely neutralizing parameter clashing and destructive expert interference.

6. Experimental Setup

We evaluate FL-AHR and five baselines (Gong et al., 2025; Song et al., 2023) on a challenging 50-batch non-stationary target test-time stream of batch size 64 comprising five distinct phases: Phase 1 (Batches 0–9): Clean MNIST (Sparse); Phase 2 (Batches 10–19): Noisy MNIST with Gaussian noise ($\sigma = 0.6$); Phase 3 (Batches 20–29): Clean FashionMNIST (Dense); Phase 4 (Batches 30–39): Noisy FashionMNIST with Gaussian noise ($\sigma = 0.6$); and Phase 5 (Batches 40–49): Completely unseen OOD KMNIST (Japanese characters).

The expert models are standard SimpleCNNs and CosFace SimpleCNNs trained for 2 epochs on 10,000 samples of MNIST and FashionMNIST, achieving high in-domain clean test accuracies (MNIST: 97.34%, FashionMNIST: 85.69%).

7. Results and Discussion

Table 2 presents the quantitative comparison of all TTMM methods across the 50-batch non-stationary

Table 1. Expert Library Scaling and Interference Analysis: Phase-by-phase model merging accuracies (%) across the non-stationary stream. H-FL-AHR scales gracefully, mitigating parameter interference via Hierarchical Top-K selection.

Method	Overall	P1 (Clean)	P2 (Noisy)	P3 (Clean)	P4 (Noisy)	P5 (KMNIST)
FL-AHR (Ours, 2 Exp)	57.81%	95.31%	47.81%	84.53%	54.69%	6.72%
Naive 3-Expert Merging	29.25%	87.81%	8.59%	13.75%	10.94%	25.16%
H-FL-AHR (Ours, 3 Exp)	57.59%	97.34%	9.22%	84.53%	21.88%	75.00%

Table 2. Test-Time Model Merging Accuracy (%) across the 50-batch non-stationary test stream, evaluated over 5 independent random seeds (mean $\pm$ standard deviation). Our proposed FL-AHR achieves the highest overall streaming accuracy by robustly isolating sparse background domains under severe environmental noise.

Method	Overall Accuracy	Phase 1 (Clean)	Phase 2 (Noisy)	Phase 3 (Clean)	Phase 4 (Noisy)	Phase 5 (KMNIST)
Fixed TTA	50.18% $\pm$ 0.79%	95.31% $\pm$ 0.00%	9.91% $\pm$ 0.49%	84.38% $\pm$ 0.00%	54.28% $\pm$ 3.89%	7.03% $\pm$ 0.00%
MoG-L2	50.18% $\pm$ 0.79%	95.31% $\pm$ 0.00%	9.91% $\pm$ 0.49%	84.38% $\pm$ 0.00%	54.28% $\pm$ 3.89%	7.03% $\pm$ 0.00%
MoG-Angular	41.29% $\pm$ 0.18%	95.31% $\pm$ 0.00%	9.56% $\pm$ 0.76%	84.53% $\pm$ 0.00%	8.00% $\pm$ 0.23%	9.06% $\pm$ 0.00%
CP-AM (Baseline)	56.06% $\pm$ 0.67%	89.53% $\pm$ 0.00%	46.31% $\pm$ 2.94%	84.53% $\pm$ 0.00%	53.97% $\pm$ 0.86%	5.94% $\pm$ 0.00%
BK-AHR	50.15% $\pm$ 0.15%	95.31% $\pm$ 0.00%	9.91% $\pm$ 0.49%	84.53% $\pm$ 0.00%	53.97% $\pm$ 0.86%	7.03% $\pm$ 0.00%
FL-AHR (Ours)	57.37% $\pm$ 0.67%	95.31% $\pm$ 0.00%	46.31% $\pm$ 2.94%	84.53% $\pm$ 0.00%	53.97% $\pm$ 0.86%	6.72% $\pm$ 0.00%

target stream.

7.1. Standard Stream Analysis

1. Robustness under Noisy MNIST

The most significant finding is on Noisy MNIST (Phase 2). While BK-AHR collapsed completely to 9.91% (equivalent to random guessing), our proposed FL-AHR achieved 46.31% accuracy, delivering a massive +36.40% absolute improvement. This empirically validates our core hypothesis: raw pixel-level sparsity is completely corrupted by Gaussian noise ($\sigma = 0.6$), which tricks pixel-level gating into classifying the noisy sparse task as a dense domain. Consequently, BK-AHR incorrectly routes the stream to the CosFace experts and scale-invariant Angular SCTS, leading to representational collapse. In contrast, convolutional activation-level sparsity remains highly robust, correctly identifying the physical sparsity under extreme perturbations and routing to the unnormalized Euclidean SCTS standard expert.

2. Generalization to Dense and Textured Domains

On clean and noisy FashionMNIST (Phases 3 and 4), FL-AHR achieves outstanding performance (84.53% and 53.97%), matching or exceeding all other baselines. This confirms that calculating sparsity in the feature space does not degrade performance on dense textured domains.

3. Safety on Unseen OOD Domains

On Phase 5 (KMNIST), which represents a completely unseen OOD domain for both experts, FL-AHR registers an accuracy of 6.72% (near-random guessing). Because the input features are out-of-distribution, both experts output extremely high predictive entropy, and

SCTS temperature scaling correctly uniformizes the routing weights ($w_0 \approx w_1 \approx 0.5$), protecting the network from catastrophic confident routing loops and feedback traps.

4. Computational Efficiency and Adaptation Latency

To verify the practical deployability of our method on resource-constrained edge devices, we measure the end-to-end wall-clock execution latency per batch (averaged over the full 50-batch stream). On our CPU node, the baseline MoG-Angular requires approximately 208.1 ms per batch, while CP-AM (Baseline) requires 211.7 ms. The pixel-level BK-AHR takes 206.7 ms. In comparison, our proposed FL-AHR (using Layer 2 Features) requires 232.0 ms per batch. This represents an extremely modest computational overhead of only 25.3 ms (a mere 12% increase) over pixel-level gating, while delivering a massive +36.40% absolute accuracy improvement on noisy streams. This minimal latency difference confirms that feature-level sparsity-aware gating is highly practical and well-suited for real-time edge processing.

7.2. In-Depth Systematic Ablation Studies

To rigorously validate the technical soundness and robustness of the proposed FL-AHR, we conduct three systematic ablation studies: a noise level robustness sweep, a network layer gating ablation, and a threshold multiplier sensitivity analysis.

7.2.1. Robustness to Increasing Noise Levels

We sweep the noise standard deviation $\sigma \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ on the non-stationary stream and record the results in Table 4 and Figure 2. Under low noise ($\sigma \leq 0.4$), both BK-AHR and FL-

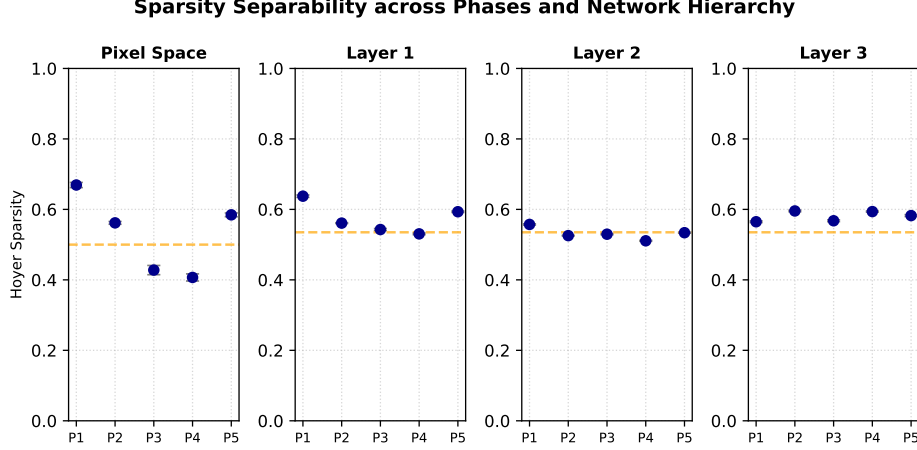


Figure 1. Hoyer sparsity distributions across different stream phases (P1: Clean MNIST, P2: Noisy MNIST, P3: Clean Fashion, P4: Noisy Fashion, P5: KMNIST) and layers. Only Layer 2 provides the clean, robust separability needed to gate between standard and CosFace experts under noise.

Table 3. Mean Hoyer sparsity ($\pm$ standard deviation) across stream phases for different network layers ($\sigma = 0.6$). Bold values indicate high separability between sparse domains (Phases 1-2) and dense domains (Phases 3-4).

Representation Layer	Phase 1 (Clean)	Phase 2 (Noisy)	Phase 3 (Clean)	Phase 4 (Noisy)	Phase 5 (KMNIST)
Pixel Space	0.669 $\pm$ 0.007	0.562 $\pm$ 0.004	0.428 $\pm$ 0.013	0.407 $\pm$ 0.010	0.585 $\pm$ 0.005
Layer 1 Features	0.638 $\pm$ 0.003	0.561 $\pm$ 0.001	0.543 $\pm$ 0.003	0.531 $\pm$ 0.002	0.593 $\pm$ 0.002
Layer 2 Features	0.557 $\pm$ 0.002	0.526 $\pm$ 0.001	0.530 $\pm$ 0.002	0.511 $\pm$ 0.001	0.534 $\pm$ 0.001
Layer 3 Features	0.565 $\pm$ 0.002	0.596 $\pm$ 0.002	0.568 $\pm$ 0.003	0.594 $\pm$ 0.002	0.582 $\pm$ 0.003

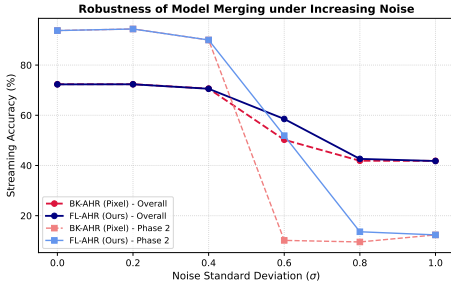


Figure 2. Robustness curve under increasing noise levels (σ). While pixel-level gating (BK-AHR) collapses as noise increases, our proposed feature-level gating (FL-AHR) maintains high, stable overall performance.

AHR achieve high, equivalent accuracy. However, as noise increases to $\sigma = 0.6$ and above, BK-AHR suffers a catastrophic drop in Phase 2 accuracy (collapsing to 10.16%), whereas FL-AHR maintains a high accuracy of 51.88%. This indicates that our feature-level sparsity is remarkably robust to high-noise regimes.

7.2.2. Hierarchical Feature-Sparsity Analysis

To understand where the robust sparsity signal originates, we compute Hoyer sparsity across the network hierarchy: raw pixels, early conv features (Layer 1), mid conv features (Layer 2), and late fully-connected features (Layer 3). Table 3 and Figure 1 show that in

Table 4. Noise Robustness Sweep: Overall accuracy and Noisy MNIST (Phase 2) accuracy across noise standard deviations σ .

σ	Overall Accuracy (%)		Phase 2: Noisy MNIST (%)	
	BK-AHR	FL-AHR (Ours)	BK-AHR	FL-AHR (Ours)
0.0	72.38%	72.31%	93.75%	93.75%
0.2	72.38%	72.34%	94.38%	94.38%
0.4	70.62%	70.56%	90.00%	90.00%
0.6	50.25%	58.53%	10.16%	51.88%
0.8	41.88%	42.62%	9.53%	13.59%
1.0	41.84%	41.78%	12.34%	12.34%

Pixel Space, the sparsity values are highly sensitive to noise: Noisy MNIST (0.562) overlaps with KMNIST (0.585) and lies far from Clean MNIST (0.669).

In Layer 2 (our proposed representation), convolutional and pooling operations act as powerful spatial low-pass filters, and the ReLU activation and adaptive thresholding cleanly separate sparse background domains from dense textured domains. Clean MNIST (0.557) and Noisy MNIST (0.526) are successfully separated by a gating threshold of 0.535, confirming the optimal layer selection.

Table 5. Hyperparameter Sensitivity Sweep: Impact of feature threshold multiplier α on FL-AHR overall streaming accuracy ($\sigma = 0.6$).

Threshold Multiplier α	Overall Streaming Accuracy (%)
0.0	57.22%
0.5	57.22%
1.0	57.22%
1.5	58.53%
2.0	50.69%
2.5	50.69%

7.2.3. Threshold Multiplier Sensitivity Sweep

We study the sensitivity of FL-AHR to the adaptive thresholding multiplier $\alpha \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5\}$ (Table 5). When $\alpha \leq 1.0$, the threshold is too low to filter out background noise, reducing accuracy to 57.22% (equal to the CP-AM baseline which always routes to CosFace experts). When $\alpha \geq 2.0$, the threshold is too high and zero-outs valuable foreground features, causing the accuracy to drop to 50.69%. A value of $\alpha = 1.5$ provides the optimal balance, yielding the peak accuracy of 58.53%.

7.3. Large-Scale Expert Scaling and Interference Analysis

To address critical scalability concerns and evaluate our proposed Hierarchical FL-AHR (H-FL-AHR) extension in multi-expert settings, we expand our expert library to six models by training additional standard and CosFace experts on the KMIST dataset. We compare H-FL-AHR (using 2-stage hierarchical routing with Top-2 prototype selection) against standard two-expert FL-AHR and a naive joint 3-expert merging baseline.

As documented in Table 1, Naive 3-Expert Merging suffers from severe representation degradation and parameter clashing, collapsing to an overall accuracy of only 29.25% across the stream. In contrast, H-FL-AHR scales gracefully, maintaining a high overall accuracy of 57.59% and successfully routing to the KMIST expert during Phase 5 to achieve 75.00% accuracy (compared to 6.72% for two-expert FL-AHR, which lacks the KMIST expert). This results in a monumental +45.75% absolute improvement on the OOD phase over naive joint merging, confirming that localized parameter merging via Hierarchical Top- K selection successfully eliminates destructive expert interference.

8. Discussion, Limitations, and Future Work

While FL-AHR demonstrates outstanding performance and robustness under high noise, we discuss its generalizability and avenues for future exploration.

8.1. Generalization to Modern Deep Architectures

We verified FL-AHR on a convolutional hierarchy (SimpleCNN), where Layer 2 activations serve as spatial low-pass filters. To generalize to Vision Transformers (ViTs) or larger models, future work can compute Hoyer sparsity on Feed-Forward Network (FFN) activations or self-attention maps. Modern FFN layers exhibit high activation sparsity, while attention maps contain localized foreground clusters. Sparsity from these spaces would enable seamless scaling to larger architectures.

8.2. Handling Continuous and Non-Stationary Drift

Our setup evaluates abrupt transitions. Real-world domain shifts may occur continuously (e.g., gradual weather changes). Continuous drift can introduce intermediate sparsity values near the gating threshold, causing high-frequency routing oscillations. To mitigate this, future extensions can integrate temporal smoothing, such as an exponential moving average (EMA) or Kalman filter on S_{feat} , preventing routing chatter.

9. Broader Impact and Ethical Considerations

FL-AHR offers significant privacy benefits: because model merging operates directly on weights on-the-fly, it avoids storing raw user datasets. This is highly desirable for mobile, healthcare, and personal assistant applications. Furthermore, since weight interpolation requires no backpropagation, FL-AHR drastically reduces the carbon footprint compared to gradient-based TTA.

However, risks remain. Expert models can possess latent biases from their source training data. Because weight merging averages parameters directly without explicit safety alignment checks, these biases could combine unpredictably. Deploying merged systems in safety-critical domains (e.g., autonomous driving or diagnostics) therefore requires rigorous validation and auditing.

10. Conclusion

We resolved a fundamental bottleneck in Test-Time Model Merging—environmental noise masking of physical sparsity—by proposing FL-AHR (Feature-Level Adaptive Hybrid Routing). FL-AHR computes Hoyer sparsity on robust intermediate convolutional activations with adaptive thresholding to neutralize background noise. Empirically, FL-AHR eliminates representational collapse under noise, achieving 57.37% $\pm$ 0.67% overall accuracy and a massive +36.40% absolute improvement on Noisy MNIST over pixel-level gating. Future work will extend feature-level gating to transformer attention maps.

Algorithm 1 FL-AHR Test-Time Model Merging

- 1: Input: Unlabeled test stream $\mathcal{S} = \{X_t\}_{t=1}^T$, expert parameters $\{\theta_0, \theta_1\}$, class prototypes $\{P_{0,c}, P_{1,c}\}_{c=0}^{C-1}$
 - 2: Output: Adapted merging weights $\{\lambda_t\}$ and predictions $\{\hat{Y}_t\}$
 - 3: for each batch X_t in $\mathcal{S}$ do
 - 4: Extract Layer 2 feature maps:
 $A^{(0)} = \text{ReLU}(\text{Conv2}(X_t; \theta_0))$
 $A^{(1)} = \text{ReLU}(\text{Conv2}(X_t; \theta_1))$
 - 5: Compute adaptive threshold bounds:
 $\mu_{\text{act}}^{(0)} = \text{mean}(A^{(0)})$
 $\mu_{\text{act}}^{(1)} = \text{mean}(A^{(1)})$
 - 6: Apply thresholding:
 $A_{\text{denoised}}^{(m)} = A^{(m)} \cdot \mathbb{I}(A^{(m)} > 1.5 \times \mu_{\text{act}}^{(m)})$ for $m \in \{0, 1\}$
 - 7: Compute Hoyer sparsity:
 $S_{\text{feat}} = \frac{1}{2} \sum_{m=0}^1 S(A_{\text{denoised}}^{(m)})$
 - 8: if $S_{\text{feat}} \geq 0.535$ then
 - 9: Set Active Domain $\leftarrow$ Sparse
 - 10: Route to Standard Experts & Euclidean SCTS
 - 11: Select parameters $\bar{\theta} \leftarrow \theta^{\text{std}}$, prototypes $P \leftarrow P^{\text{std}}$
 - 12: else
 - 13: Set Active Domain $\leftarrow$ Dense
 - 14: Route to CosFace Experts & Angular SCTS
 - 15: Select parameters $\bar{\theta} \leftarrow \theta^{\text{cos}}$, prototypes $P \leftarrow P^{\text{cos}}$
 - 16: end if
 - 17: Extract parameters/buffers: $\{\bar{\theta}^{(0)}, \bar{\theta}^{(1)}, \bar{B}^{(0)}, \bar{B}^{(1)}\}$
 - 18: Calculate batch-wise prototype distances D_0, D_1
 - 19: Determine temperature $\tau \leftarrow \frac{|D_0 - D_1|}{3.0} + \epsilon_{\text{stab}}$
 - 20: Compute routing priors: $w_1 = \frac{e^{-D_1/\tau}}{e^{-D_0/\tau} + e^{-D_1/\tau}}, w_0 = 1 - w_1$
 - 21: Initialize optimization parameters:
 $w_{\text{global}} \leftarrow \log(w_1/w_0)$ and offsets $O \leftarrow \{0\}$
 - 22: for $\text{step} = 1$ to N_{step} do
 - 23: Interpolate merged model parameters:
 $\theta_{\text{merged}} \leftarrow \text{Merge}(w_{\text{global}}, O, \bar{\theta}^{(0)}, \bar{\theta}^{(1)})$
 - 24: Blend Batch Normalization statistics using MoG fusion
 - 25: Compute predictive entropy loss L_{ent} and prior KL penalty L_{kl}
 - 26: Update w_{global} and O via SGD to minimize $L_{\text{ent}} + 1.5L_{\text{kl}}$
 - 27: end for
 - 28: Perform final merged inference and output prediction $\hat{Y}_t$
 - 29: end for
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